## Network Topology & Significance

- Total Input Proteins: 44 (Naba Matrisome)
- Number of Identified Edges: 35
- Expected Number of Edges: 5 (Random Null)
- Average Node Degree: 1.59 connections/node
- Avg. Local Clustering Coefficient: 0.367
- PPI Enrichment P-value: $P < 1.0 \times 10^{-16}$
  (Significantly more interactions than expected, confirming biologically coordinated network)

## STRING Protein-Protein Interaction (PPI) Network
## 44 Peritoneal Effluent Matrisomal DEGs

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### Interaction Evidence Legend

- Gene Neighborhood (Chromosomal proximity)
- Gene Fusions (Observed fusion events)
- Gene Co-occurrence (Phylogenetic profiles)
- Experimentally Determined Interactions
- Curated Databases (KEGG, Reactome, etc.)
- Automated Textmining (PubMed abstracts)
- Co-expression (Conserved RNA expression)
- Protein Homology (Transferred from orthologs)

## Key Network Hubs by Degree

**• CCL5** **(Degree = 14)**

*Pro-inflammatory chemokine; central exudate hub*

**• TNFAIP6** **(Degree = 6)**

*Matrix hyaluronan-binding glycoprotein*

**• LTA / LTB** **(Degree = 5)**

*Lymphotoxin inflammatory cytokines*

**• C1QA / C1QB** **(Degree = 5)**

*Classical complement subcomponents in ECM*

**• COL8A1 / COL9A2** **(Degree = 4)**

*Core structural fibrillar/network collagens*

**• SPP1 (Osteopontin)** **(Degree = 4)**

*Macrophage-secreted matricellular driver*

**• ADAM10 / ADAM19** **(Degree = 3)**

*Peritoneal membrane sheddases & metalloproteases*